

Reinforcement Learning

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Examples

AlphaGo (2016), DeepMind



Mastering the game of Go with deep neural networks and tree search

David Silver, Aja Huang, et al.

Nature, 2016, Jan. 28

Citations

AlphaGo Zero (2017), DeepMind

Mastering the game of Go without human knowledge

Silver, D., et al.

Nature, 2017, Oct. 19

Demostration

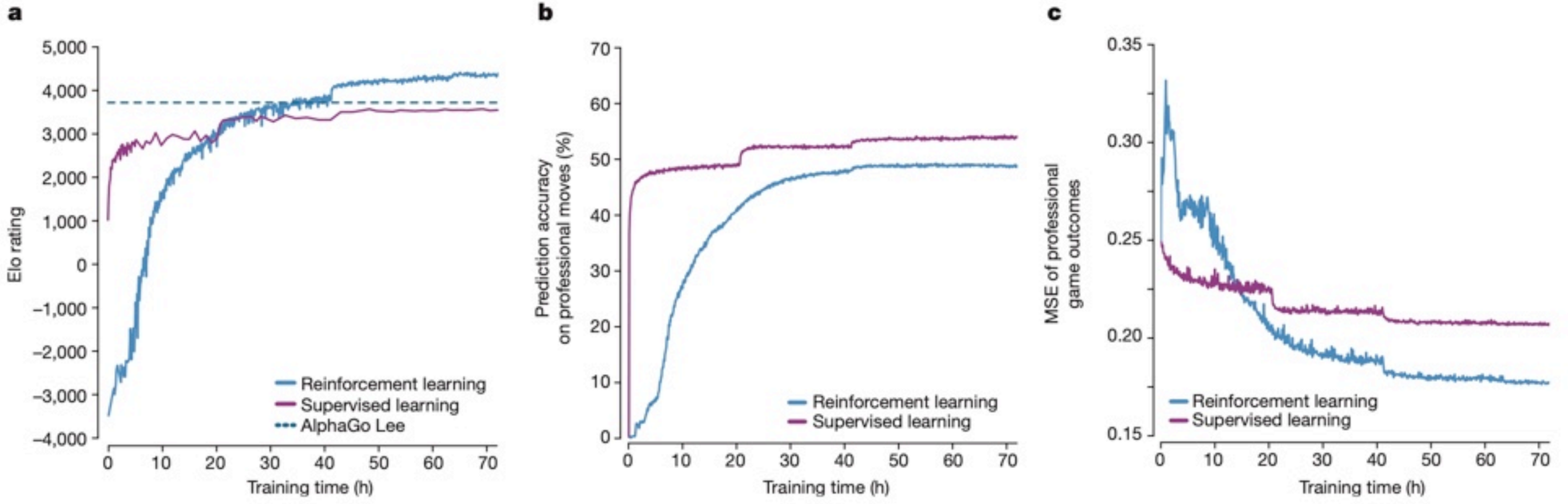


Figure 3 | Empirical evaluation of AlphaGo Zero. a, Performance of self-play reinforcement learning. The plot shows the performance of each MCTS player α_{θ_i} from each iteration i of reinforcement learning in AlphaGo Zero. Elo ratings were computed from evaluation games between different players, using 0.4 s of thinking time per move (see Methods). For comparison, a similar player trained by supervised learning from human data, using the KGS dataset, is also shown. **b**, Prediction accuracy on human professional moves. The plot shows the accuracy of the neural network f_{θ_i}, at each iteration of self-play i, in predicting human professional moves from the GoKifu dataset. The accuracy measures the percentage of positions in which the neural network assigns the highest probability to the human move. The accuracy of a neural network trained by supervised learning is also shown. **c**, Mean-squared error (MSE) of human professional game outcomes. The plot shows the MSE of the neural network f_{θ_i}, at each iteration of self-play i, in predicting the outcome of human professional games from the GoKifu dataset. The MSE is between the actual outcome $z \in \{-1, +1\}$ and the neural network value v, scaled by a factor of $\frac{1}{4}$ to the range of 0–1. The MSE of a neural network trained by supervised learning is also shown.

Alpha Zero (2017), DeepMind

Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm

Silver, D., et al.

arXiv.org, 2017, Dec. 5

Citation

MuZero (2020), DeepMind

Mastering Atari, Go, chess and shogi by planning with a learned model

Schrittwieser, …, Silver, D.

Nature, 2020, Dec. 23

Citations

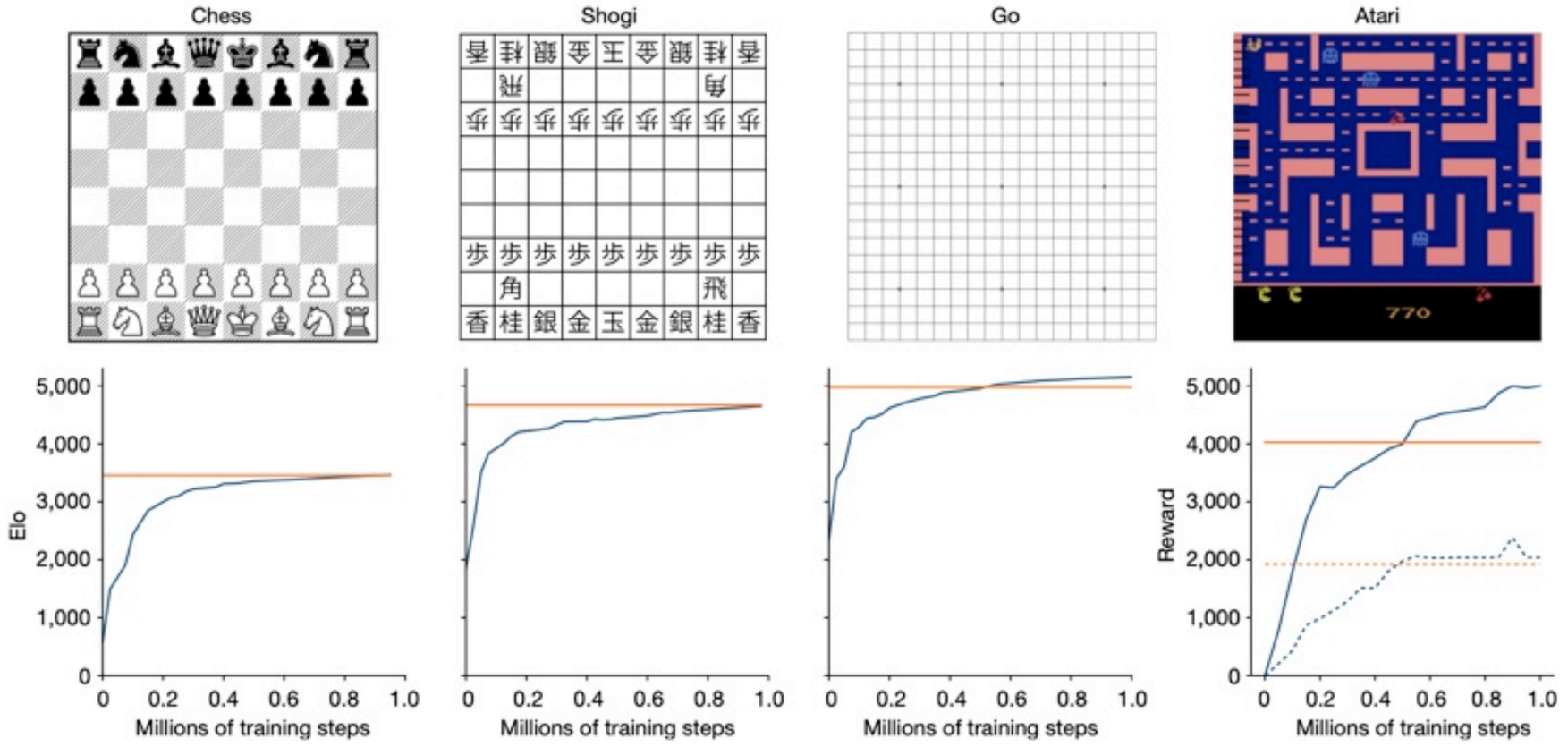


Fig. 2 | Evaluation of MuZero throughout training in chess, shogi, Go and Atari. The *x* axis shows millions of training steps. For chess, shogi and Go, the *y* axis shows Elo rating, established by playing games against AlphaZero using 800 simulations per move for both players. MuZero's Elo is indicated by the blue line and AlphaZero's Elo is indicated by the horizontal orange line. For Atari, mean (full line) and median (dashed line) human normalized scores across all 57 games are shown on the *y* axis. The scores for R2D2[19] (the previous state of the art in this domain, based on model-free RL) are indicated by the horizontal orange lines. Performance in Atari was evaluated using 50 simulations every fourth time step, and then repeating the chosen action four times, as in previous work[39]. Supplementary Fig. 1 studies the repeatability of training in Atari.

Atari (2015), DeepMind

Human-level control through deep reinforcement learning

Mnih, V., Kavukcuoglu, K., Silver, D., et al.

Nature, 2015, Feb. 25

Beating humans in most of 49 Atari games



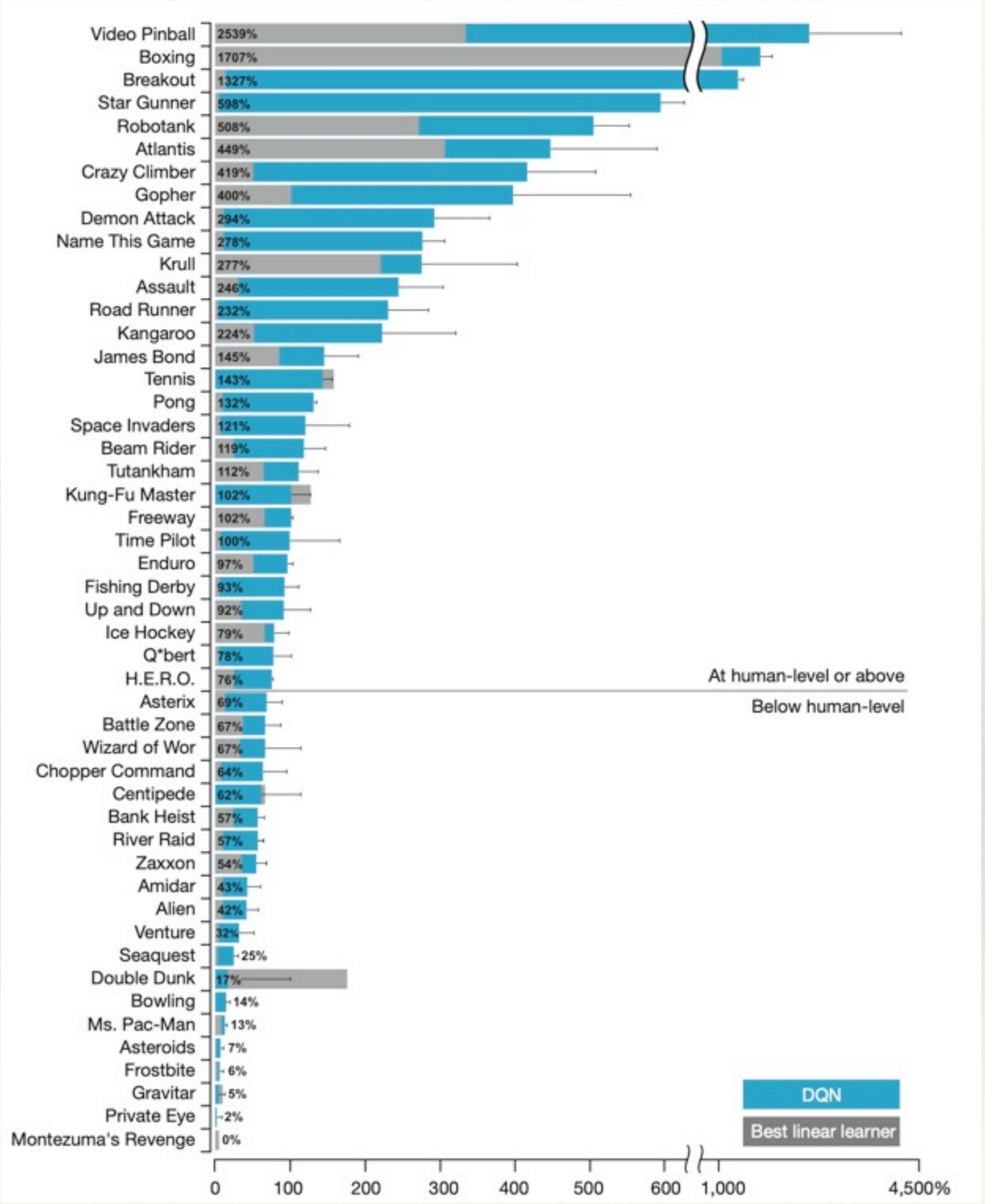


Figure 3 | Comparison of the DQN agent with the best reinforcement learning methods[15] in the literature. The performance of DQN is normalized with respect to a professional human games tester (that is, 100% level) and random play (that is, 0% level). Note that the normalized performance of DQN, expressed as a percentage, is calculated as: 100 × (DQN score − random play score)/(human score − random play score). It can be seen that DQN outperforms competing methods (also see Extended Data Table 2) in almost all the games, and performs at a level that is broadly comparable with or superior to a professional human games tester (that is, operationalized as a level of 75% or above) in the majority of games. Audio output was disabled for both human players and agents. Error bars indicate s.d. across the 30 evaluation episodes, starting with different initial conditions.

2017, Libratus in Texas Hold'em (德州扑克), CMU

Libratus_CMU



Libratus: Superhuman AI for heads-up no-limit poker

Noam Brown & Tuomas Sandholm

IJCAI, 2017

Citations

2020, June 21, StarCraft, inspire.ai

星际争霸， 启元世界



Other applications

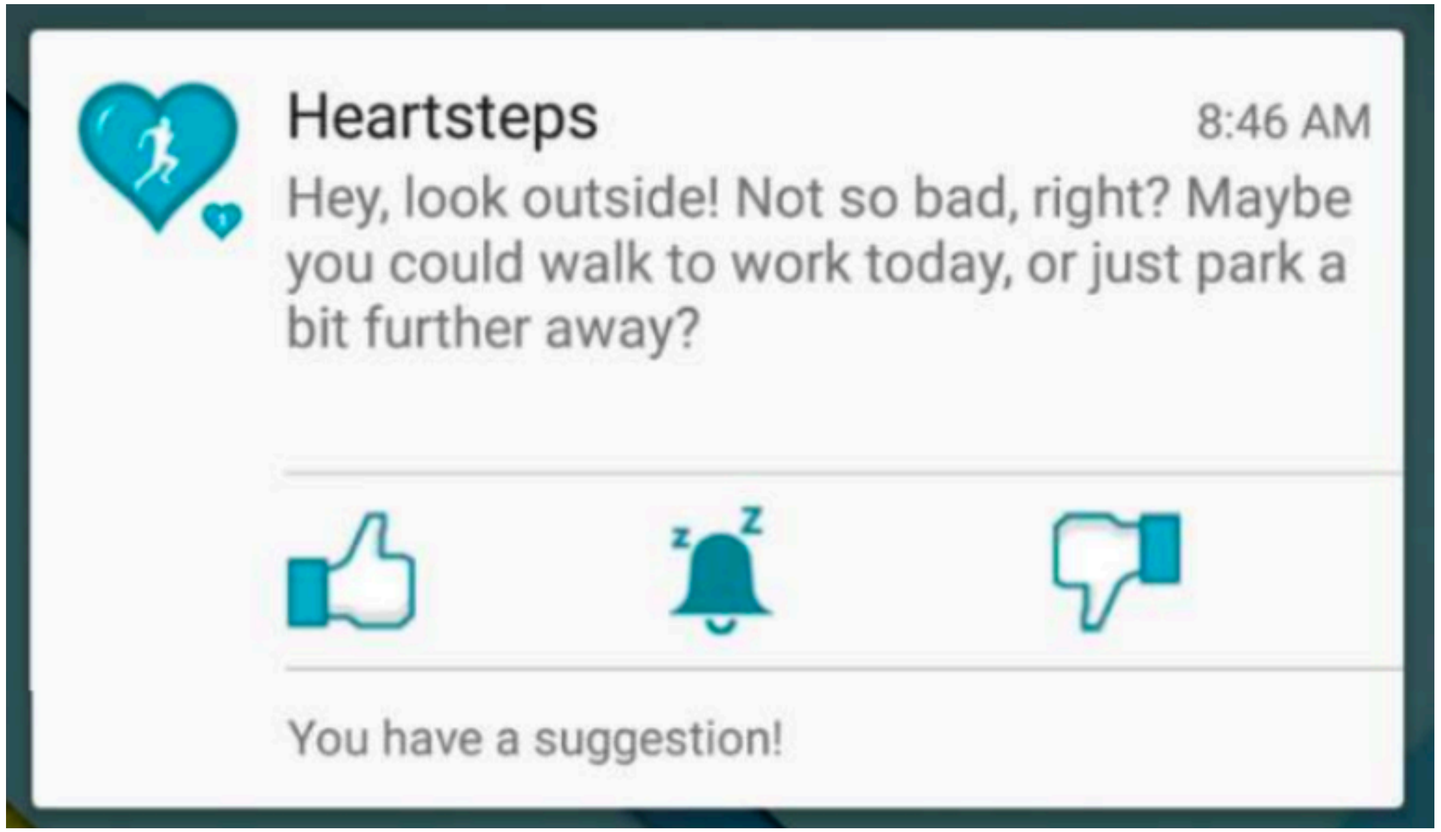
Robotics





Chelsea Finn, Sergey Levine, Pieter Abbeel, 2015

Health



Personalized HeartSteps: A Reinforcement Learning Algorithm for Optimizing Physical Activity.

Liao, Greenwald, Klasnja, Murphy, 2019

And many more ……

References

- Sutton and Barto: Reinforcement Learning
- David Silver's lecture notes
- Stanford CS234

